Introductions

COGS 108 Spring 2026
Jessica Cheng
D1

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OH: Tue 11:00am - 1:00pm

Discussion slides and materials adapted from Sam Lau (TA: WI20)

About Me

- Jessica Cheng
- MS in CSE, BS in CS at UCSD
- Email: becheng@ucsd.edu
- Office Hour: Tue 11:00am - 1:00pm
- https://ucsd.zoom.us/j/7856425053

Section Philosophy

- Attendance is **required for quizzes**
- Each section will:
  - 5–10 min review of previous material
  - ~20 min quiz on previous week
  - Help with labs, assignments, and questions
- Week 1:
  - Python review only
  - No quiz or submissions

Weekly Quizzes in section

- Goals:
  - To help you keep on top of the material covered in lecture
  - To provide an AI-free assessment of what you actually learned
- How:
  - Bring your laptop or tablet
  - On Canvas **inside LockDown Browser** with access code
  - **Cannot be taken outside of section**
  - TA will proctor, **NO AI** or other cheating!!
  - ~10-20 questions, mostly easy multiple choice, one or two thinking or coding problems
- Additional Note:
  - Not your assigned section? **Inform your TA** before quiz starts to get access.

LockDown Browser Requirement

- You must use LockDown Browser to take the quiz
- Install it **BEFORE** coming to section (no time during quiz)
- Available here: https://tinyurl.com/3w7jt3dp
- **Test it in advance** to avoid technical issues
- Important:
  - No installation = cannot take the quiz on time
  - Bring a charged laptop

SCAN ME

Project

- Form a group of 4-5 students
- Feel free to talk to others right now! Chat with your classmates about your interests, region, skills etc.
- Use EdStem
- Start working towards the project as soon as possible
- https://github.com/COGS108/Projects

Programming

This course assumes basic programming knowledge

- But not much!

Programming

Resources:

- Codeacademy
- Start Here: https://github.com/COGS108/Tutorials/blob/master/01-Python.ipynb
- Python in detail: https://jakevdp.github.io/PythonDataScienceHandbook/
- Pandas: https://www.dataschool.io/python-pandas-tips-and-tricks/
- Git: https://guides.github.com/activities/hello-world/

Programming

Cheatsheets

- Google: ‘python cheatsheet’, ‘pandas cheatsheet’, ‘git cheatsheet’ (find one that’s good for you)

Policy on AI Generated Content

AI Use Permitted with Citation (and caution!)

Class policy on generative AI

AI tools may be used freely in this course provided their use is transparent and properly cited. All AI contributions must be credited just as any other source would be cited. Students are fully responsible for the accuracy, originality, and quality of all submitted work. This expectation is in place to ensure that all submitted work reflects the student's own learning and skills. Suspected submission of AI-generated content without citation or presenting it as entirely one's own work will be reported to the Academic Integrity Office in accordance with university policy.

AI should be used as a tutor and teammate. It should not be used to do your work for you. More assistance from AI is acceptable on projects, less on labs, and very little and only conceptual help should be used on assignments.

When used for writing, it works best if you talk to it about ideas and then pick your own words, then use it again to edit your draft, then make sure you read and make final edits.

When used for programming, as a beginner talk to AI about ideas for solving the problem, ask it to explain concepts and syntax and functions. Then write the code yourself. Ask AI to help you understand errors, but you need to struggle with solving it yourself in order to learn. Ask AI to generate practice problems for you to solve.

When used for programming as an expert, then you write a design document of how to solve the problem and allow AI to work on parts of the design document. AI generate and hand validate unit tests. Always read the code AI generated, ask AI to critique the implementation and explain any errors.

Git

Version control system!

- Go to https://git-scm.com/downloads
- Choose your Operating System (Windows/OS X/Linux)
- Follow the steps specific to your OS
- Verify installation: In terminal type “git –-version”

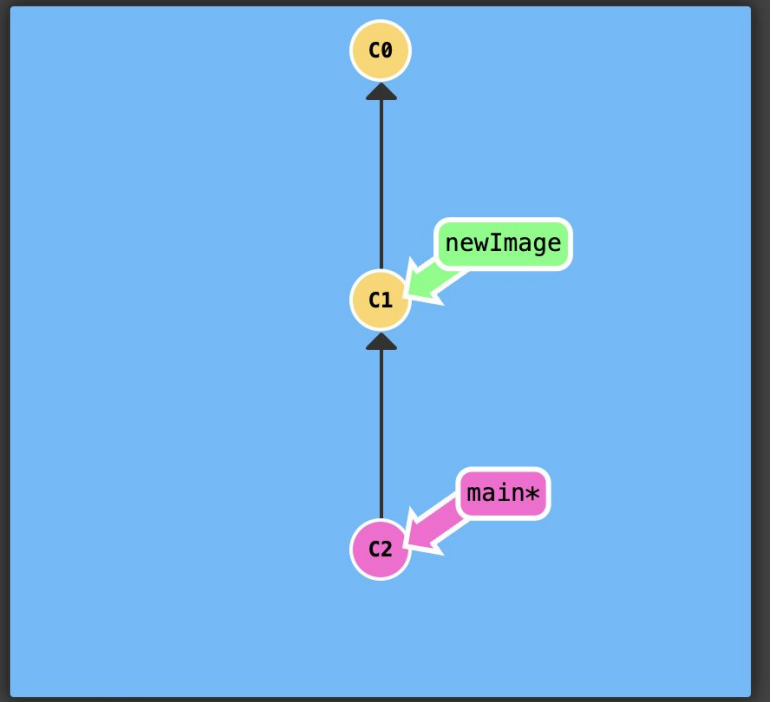
learngitbranching.js.org

Git Demonstration

Let's try to put some work on this new branch. Hit the button below.

git commit

Oh no! The `main` branch moved but the `newImage` branch didn't! That's because we weren't "on" the new branch, which is why the asterisk (*) was on `main`.



https://about.gitlab.com/images/press/git-cheat-sheet.pdf

A Git installation

For GNU/Linux distributions, Git should be available in the standard system repository. For example, in Debian/Ubuntu please type in the **terminal**:

```
$ sudo apt-get install git
```

If you need to install Git from source, you can get it from git-scm.com/downloads.

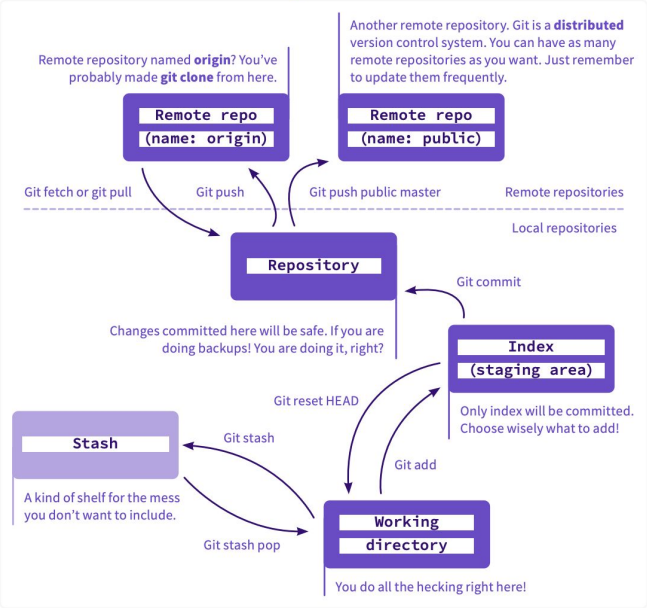
An excellent Git course can be found in the great **Pro Git** book by Scott Chacon and Ben Straub. The book is available online for free at git-scm.com/book.

B Ignoring Files

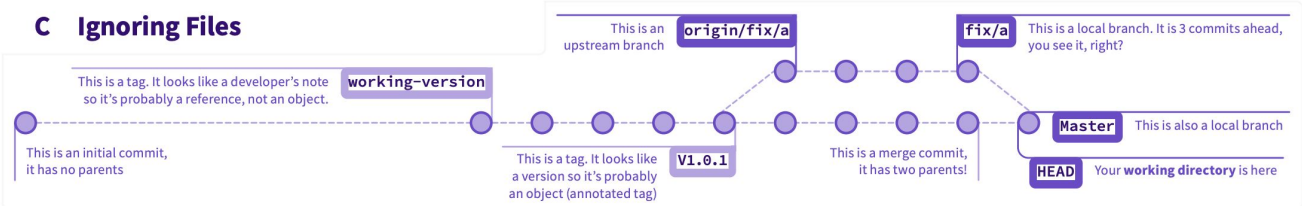
```
$ cat .gitignore
/logs/*
!logs/.gitkeep
/tmp
*.swp
```

Verify the .gitignore file exists in your project and ignore certain type of files, such as all files in **logs** directory (excluding the **.gitkeep** file), whole **tmp** directory and all files ***.swp**. File ignoring will work for the directory (and children directories) where **.gitignore** file is placed.

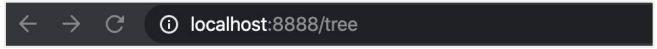
D The zoo of working areas



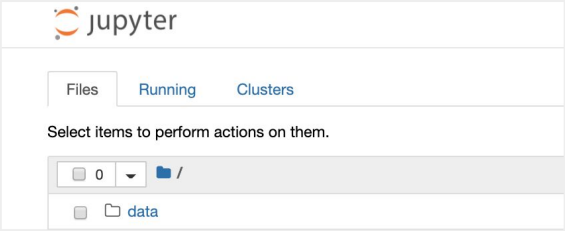
C Ignoring Files

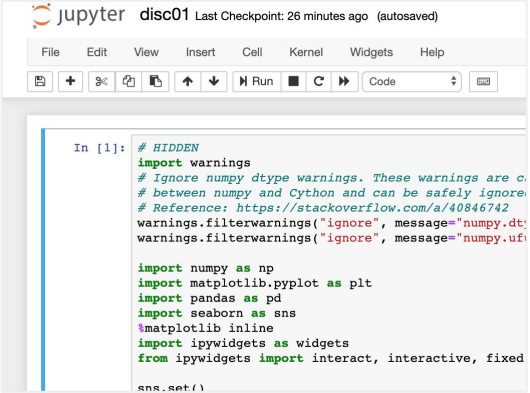


Jupyter



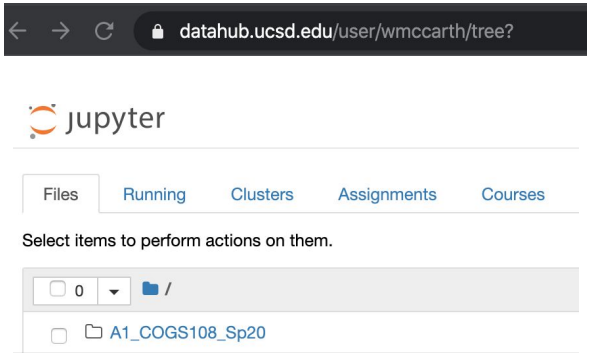
- Python code is run on a python interpreter
- Jupyter is a program that creates an interface for typing python code in a browser, that also runs that code in a python interpreter
- What does this mean?!
  - Jupyter is a way of running python programs from a browser (like chrome) (hooray!)





datahub.ucsd.edu

- Jupyter runs python code in a browser.
  - But Jupyter is itself just a program that's running on a computer somewhere.
- datahub lets you interact with Jupyter that's running somewhere else.



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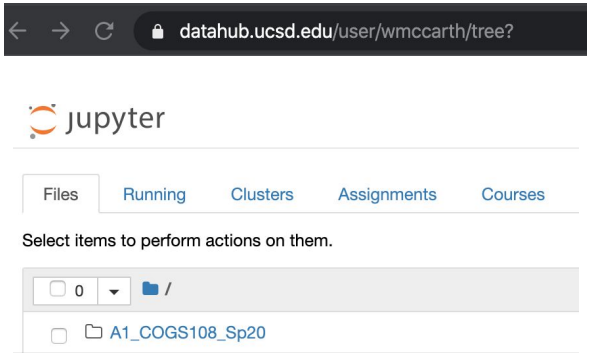
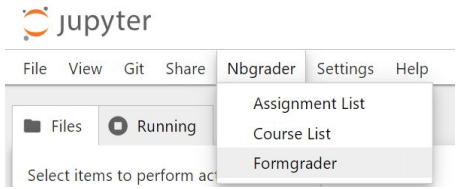
- What does this mean?!
  - You don’t need to worry about installing Jupyter
  - You can use datahub to create and run python programs (online)
  - You can use this interface to fetch and submit assignments



Working on your assignments

- Log into datahub.ucsd.edu
- Go to Assignments tab (or Nbgrader->Assignment List if you are using the new container)
- ‘fetch’ assignments you have access to -> Submit after completion
- Demo of this workflow



Python Review Notebook

https://datahub.ucsd.edu/hub/home